Akira Murakami

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a-uuu-a.github.io/portfolio/en/ Google Scholar

I am a doctoral student at the ICD Lab, Graduate School of Information Sciences, Tohoku University, advised by Yoshifumi Kitamura and Kazuyuki Fujita. My research lies at the intersection of Human-Computer Interaction (HCI) and architecture. My recent work focuses on shape-changing robotic furniture and architectural systems for reconfigurable spaces, as well as spatial layout editing using AI and XR.

Education

Oct 2026 – Present	**Tohoku University** Doctoral Program in Information Science Advisors: Yoshifumi Kitamura, Kazuyuki Fujita
Oct 2024 – Sep 2026	**Tohoku University** Master of Information Sciences
Apr 2021 – Sep 2024	**Tohoku University** Bachelor of Engineering

Research Publications

Conference Papers & Abstracts · peer reviewed

[1] **Akira Murakami**, Kazuyuki Fujita, Yuki Onishi, and Yoshifumi Kitamura. OrigamiWalls: A Shape-Changing Robotic Partitioning System for Diverse Spatial Reconfigurations. *Proceedings of the 2026 Designing Interactive Systems Conference (DIS '26), Singapore, pp. 4559–4576, 2026*

[2] Ryohei Miyadera, **Akira Murakami**, Nazuki Terakawa, Keito Tanemura, Mao Fujino, Hisayoshi Sakahira, Masanori Fukui. Mathematical Games of Dice - A Research Based Calculations Using Computer Algebra Systems. *The Proceedings of the 5th International Conference on Management, Engineering, Science, Social Science and Humanities, Penang, pp.54, 2020*

[3] Ryohei Miyadera, Masanori Fukui, Nazuki Terakawa, Keito Tanemura, **Akira Murakami**, Mariko Kashiwagi, Ryuhei Utsuhara. A Variant of Bogus Nim - Experimental Mathematics Approach. *The Proceedings of the 5th International Conference on Management, Engineering, Science, Social Science and Humanities, Penang, pp.55, 2020*

[4] Ryohei Miyadera, Mariko Kashiwagi, Masanori Fukui, **Akira Murakami**, Keito Tanemura, Itsuki Kitagawa, Kazuya Hiramatsu and Shintaro Kaiomto. Wythoff's Game with a Pass. *JCDCG^3 2019, Tokyo, pp.100-101, 2019*

[5] Ryohei Miyadera, Mariko Kashiwagi, Michitada Hayashi, Sintarou Kaimoto, Keito Tanemura, **Akira Murakami**, Ryuhei Utsuhara. "Rugby Nim" that is a Variant of "Sliding". *JCDCG^3 2018, Manila, pp. 41-43, 2018*

[6] Ryohei Miyadera, Mariko Kashiwagi, Nazuki Terakawa, Syouta Masuda, Keito Tanemura, Sintarou Kaimoto, Ryuhei Utsuhara, **Akira Murakami**. "Sliding With a Push" that is a Variant Of "Sliding". *JCDCG^3 2018, Manila, pp. 39-41, 2018*

Posters & Demonstrations · peer reviewed

[1] **Akira Murakami**, Kodai Hatakeyama, Ryo Ooka, Kazuyuki Fujita, Guanghan Zhao, and Yoshifumi Kitamura. AmbiXR: Adaptive Visual Syntax for Resolving Ambiguous Instructions in LLM-Assisted XR Authoring. *32nd ACM Symposium on Virtual Reality Software and Technology (VRST '26), Sendai, Japan, November 16–18, 2026, ACM, 2 pages*

Upcoming presentation

Journal Articles · peer reviewed

[1] Ryohei Miyadera, **Akira Murakami**, Nazuki Terakawa, Keito Tanemura, Mao Fujino and Hisayoshi Sakahira. Mathematical Games of Dice - A Research Based Calculations Using Computer Algebra Systems. *Solid State Technology, Vol. 63, No. 1s, pp. 619 - 629, 2020*

[2] Ryohei Miyadera, Nazuki Terakawa, Keito Tanemura, **Akira Murakami**, Mariko Kashiwagi, Ryuhei Utsuhara. A Variant of Bogus Nim – Experimental Mathematics Approach. *Solid State Technology, Vol. 63, No. 1s, pp. 662 – 669, 2020*

Research Publications (Japanese)

Journal Articles · peer reviewed

[1] 鈴木利菜，**村上 聡**，寺川渚月，坂本悠輔，宮寺良平，川上さくら. 面数が異なるサイコロを用いたサイコロゲームに関する研究. ゲーム学会論文誌，14-1，pp.14-17，2020

Conference Presentations & Demonstrations · non refereed

[1] 金井 美桜，藤田 和之，**村上 聡**，齋藤 五大，北村 喜文. 複数の音源対象を持つコンテンツの視聴における視線に応じた振動触覚提示手法. 第 30 回日本バーチャルリアリティ学会大会 (VRSJ'25)，2025 年 9 月，大阪

[2] 菊池 勇吾, 金井 美桜, 佐々木 涼, **村上 聡**, 伊藤 勝徳, 尾形 優斗, 柴藤 りん, 佐藤 生琉, 西山 喜尋. 豆|頁：なんかきられてももどるやつ. 第 30 回日本バーチャルリアリティ学会大会 (VRSJ'25)，2025 年 9 月，大阪

[3] **村上 聡**，藤田 和之，北村 喜文. 折紙構造を用いた自走式・形態変化パーティションデバイス. 第 32 回インタラクティブシステムとソフトウェアに関するワークショップ (WISS'24)，2024，新潟

[4] 福井 昌則，柏木 麻理子，寺本 雅治，**村上 聡**，種村 圭依人，宮寺 良平，平嶋 宗. ゲーム・パズルの変形・改良に着目した創造性発揮プロセスの実態把握.「ゲームと教育」研究部会第 12 回研究会，2019，香川

[5] 寺川 渚月，**村上 聡**，柏木 麻里子，寺本 雅治，坂本 悠輔，宮寺 良平，北川 一希，平松 和也. サイコロゲームの数学. ゲーム学会第 18 回全国大会，2019，大阪

[6] 寺川 渚月，**村上 聡**，宮寺 良平，北川 一希，平松 和也，青山 あいほ，末續 鴻輝. Wythoff のゲームの変種-制限付きの 2 つの Rook. ゲーム学会第 18 回全国大会，2019，大阪

Awards & Honors

2025	**Interverse Virtual Reality Challenge (IVRC) 2025** Project:「豆|頁：なんかきられてももどるやつ」. SEED Stage: Unity Award, CRI Middleware Award, and Audience Award. LEAP Stage: Etsuko Ichihara Award and Audience Award; one of 10 finalists from 96 submissions.
2022	**Open Hack U 2022 Spring ONLINE — Grand Prize**
2021	**Tohoku University Business Idea Contest 2021 — Grand Prize**
Feb 2020	**21st National Junior and Senior High School Web Contest — Semifinalist, High School Division**
Nov 2019	**Musashino University Mathematical Engineering Contest — Encouragement Award**
Mar 2019	**SGH Koshien — Japanese Presentation Award** Research on fostering creativity through mathematical activities and sharing these methods across Asia.
Dec 2018	**Japan Science & Engineering Challenge (JSEC) 2018 — Selected Entry**
Feb 2018	**Science Castle — Excellence Award**

Scholarships & Research Funding

Apr 2025 – Present

Graduate Program in Data Science (GP-DS), Tohoku University
200,000 JPY/month + financial support for study abroad

MiTOHOKU Program, 4th cohort — METI AKATSUKI Project
Project funding: 1,200,000 JPY

MiTOHOKU Program, 2nd cohort — METI AKATSUKI Project
Project funding: 1,000,000 JPY

Research & Development Experience

Oct 2021 – Nov 2025

Adansons, Inc.
Engineer Intern
Worked on a research project exploring skill transfer through haptic technology, from planning to developing software for exhibitions.

Apr 2021 – Apr 2023

Bridge UI, Inc.
Research Intern
Developed mobile applications for research and web applications for mathematics education.

Service

2026

ACM VRST 2026 — Student Volunteer *(upcoming)*

2025

CHI 2025 Symposium — Student Committee Member (Public Lecture / Yokohama MICE Next Generation Development Project)

2024

WISS 2024 — Student Volunteer

2019

SEAMEO Workshop — Staff
Seameo Recsam, Penang, Malaysia
Problem-Solving and Computational Thinking Skills

27–28 Aug 2018

SEAMEO Workshop — Staff
Seameo Recsam, Penang, Malaysia
Workshop on Enhancing Creativity through Transdisciplinary Mathematics Education